ENVIRONMENTAL INEQUALITY LAB

RESEARCH HIGHLIGHT

Clean Air's Benefits Reach into the Next Generation

Lessons from "The Intergenerational Effects of Early-Life Pollution Exposure" by Jonathan Colmer & John Voorheis in *Journal of Political Economy: Microeconomics*, forthcoming.

Cleaner air doesn't just help the people who breathe it — its benefits reach their children, decades later. Parents who were exposed to less air pollution before birth, thanks to the 1970 Clean Air Act, went on to have children who were more likely to attend college.

BACKGROUND

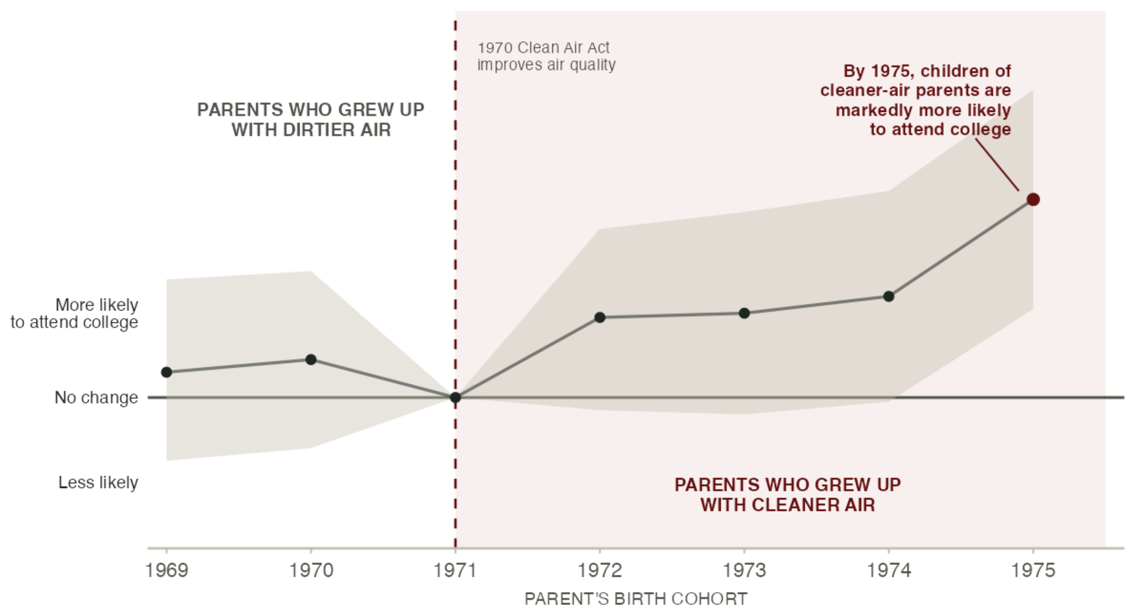
Decades of research show that **breathing polluted air early in life, even in utero, leaves lasting marks.** Children directly exposed tend to have worse health, less schooling, and lower earnings. Do those harms end there, or do they reach the next generation — the exposed children's own kids? If the effects are intergenerational, the true cost of pollution would be far larger than a single-generation accounting suggests. Whether early-life pollution carries these **intergenerational effects had never been directly measured**, because no dataset had followed families long enough to tie a parent's own early-life exposure to how their children turned out.

A new study offers the first direct evidence, drawing on a newly built dataset of more than 150 million parent–child links joined to U.S. Census survey records. The researchers focused on the **1970 Clean Air Act Amendments**, which forced hundreds of counties that exceeded federal limits to cut particulate pollution, sharply improving local air quality within a few years. That let them follow the children of people whose own prenatal exposure was shaped by the law, and ask whether cleaner air for one generation changed how the next generation fared 40 to 50 years later.

WHAT THE STUDY FOUND

CHILDREN OF PARENTS WHO GREW UP WITH CLEANER AIR WERE MORE LIKELY TO ATTEND COLLEGE

Source: Colmer & Voorheis (2025).



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Note: Each point compares one parent's birth cohort with the 1971 cohort — whether their children were more or less likely to attend college. The line is flat before 1971 and climbs after the 1970 Clean Air Act cleared the air: by the 1975 cohort, children of parents who grew up with cleaner air were about two percentage points more likely to attend college. The shaded band shows the range of uncertainty, which spanned "no change" until after the Clean Air Act took effect.

The hard part is separating the effect of cleaner air from all the other things that shape whether a child reaches college. The first instinct is to compare the children of parents who grew up in polluted places against those from cleaner ones — but the two groups differ in far more than the air they breathed, from income to schools to opportunity, any of which could explain who makes it to college. The next step is to look within the same places over time, comparing parents born just before the air cleaned up against those born just after: a child conceived a year earlier breathed an extra year of pollution. Yet birth years differ in more than air. The economy, medicine, and schooling were all shifting too, so comparing the two cohorts' children could credit cleaner air for gains those broader trends were already driving.

The researchers' way around this was to bring in a second set of counties: those whose air already met the federal limits, where the 1970 law changed nothing. Any nation-wide effects on a birth cohort's prospects showed up in these already-clean counties too. By measuring how much *more* the cleanup counties' children pulled ahead, cohort by cohort, the researchers stripped out those shared trends and isolated the effect of the air itself. This holds as long as the two kinds of counties would otherwise have moved in step — credible here, because the cleanups were forced by federal rules and national pollution levels, not by anything the families chose. If cleaner air carried no advantage to the next generation, the two groups' children should have reached college at the same rate.

That is not what the data show. **Children whose parents breathed less pollution before birth were significantly more likely to attend college.** The cleaner air delivered by the 1970 Act raised the next generation's college attendance by roughly two percentage points — about a tenth of the boost a child gets from Head Start, achieved through nothing more than cleaner air during a few months of life a generation earlier.

TAKEAWAYS

HOW THE ADVANTAGE PASSES DOWN

The effect appears to travel mostly through **parents, not genes**. When the researchers looked at adopted and step-children — who share their parents' household but not their biology — the benefit still showed up, pointing to the resources and investments a better-off parent can provide rather than any inherited biological change. Their estimates suggest heritable channels explain at most about a third of the effect.

THE FULL VALUE OF CLEANER AIR

The benefits reach a second generation, so standard analyses that stop at the people directly exposed **understate what environmental protection is worth**. Counting both generations, the lifetime earnings gains from this one improvement are 50 to 85 percent larger than the first-generation gains alone — on the scale of the mortality benefits that usually dominate these calculations.

WHOM CLEAN AIR REACHES

Low-income and disadvantaged communities breathe the most polluted air, so its hidden costs compound across generations — and cleaning the air can interrupt that cycle. This is early evidence from a single law and one outcome, and some results are suggestive rather than definitive. Even so, the central lesson is hard to miss: **the benefits of clean air outlast the people who first breathe it.**

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Colmer, Jonathan, and John Voorheis. Forthcoming. "The Intergenerational Effects of Early-Life Pollution Exposure." *Journal of Political Economy: Microeconomics*. doi.org/10.1086/740104.